

An AI **spec answer desk**, taken from pain to proven value.

Twelve stages. Four of them on architecture. Formulas only.



Samy Yusif

AI Products Section Head

Site keeps asking what the specification already answers.

Go and watch it happen. The same clause looked up by different people, in different weeks, answered from memory. Count the occurrences before you allow yourself to call it a problem.



Samy Yusif

AI Products Section Head

Size the pain before you fall in love with a solution.

$$\text{pain} = f \times t \times n \times \text{loaded rate}$$

Frequency, minutes per lookup, people involved, fully loaded rate. Then add the cost of a wrong answer times how often one slips through. If the total is small, stop here and say so out loud.



Samy Yusif

AI Products Section Head

Buy, build, or refuse. Refuse is a real answer.

Build only when the work is your differentiator and no vendor covers it. Everything else you buy or drop. Most AI failure begins as a solution walking around looking for a pain to attach to.



Samy Yusif

AI Products Section Head

Map the step you are changing, not the department.

As-is and to-be side by side. AI replaces a step inside a process, never the process. Name the exact handoff that moves and name who owns the output after it moves.



Samy Yusif

AI Products Section Head

Count the reviewer. Most business cases pretend he is free.

$$ROI = (value - cost) / cost$$

Value is hours saved times rate, plus errors avoided times cost per error. Cost is build, tokens, review time and change management. Move work from doer to checker and you moved cost, not removed it.



Samy Yusif

AI Products Section Head

Four layers, and only one of them is the model.

Ingestion that preserves clause structure. Retrieval over that structure. A generation step not allowed to leave it. A review queue for whatever comes back weak. Draw it before you cost it.



Samy Yusif

AI Products Section Head

Chunk by clause, never by character count.

```
carry doc, section, clause, revision
```

A specification is already structured. Splitting it every eight hundred characters destroys the one property that makes an answer defensible. Every chunk keeps its address in the document.



Samy Yusif

AI Products Section Head

It answers from what came back, or it says nothing.

`no clause reference = rejected`

Constrain generation to the retrieved text, require a clause reference in the output, and drop the response in code when the reference is missing. Not a polite instruction in a prompt you hope holds.



Samy Yusif

AI Products Section Head

Decide what the desk is never allowed to answer.

Contractual interpretation, variations, anything touching safety. Those route to a person by rule, not by confidence score. Write that list before launch. It is a product decision, not a technical one.



Samy Yusif

AI Products Section Head

Revisions are the risk you cannot fully mitigate.

Specifications get revised mid-project. An answer correct on revision C is wrong on D and the reader cannot see the difference. Stamp the revision on every answer, and accept some will still be read late.



Samy Yusif

AI Products Section Head

Permissions per project, full logging, a hard spend ceiling.

Who can see which project's documents. Every question and answer stored and retrievable. A monthly cap on usage-based spend, because without one a pilot turns into a board escalation.



Samy Yusif

AI Products Section Head

Set the passing score before you see any results.

```
accept if recall >= R and precision >= P
```

A labelled set built from real past questions whose answers are known. Decide which error hurts more, missing a clause or inventing one, and set the two thresholds accordingly. In writing, first.



Samy Yusif

AI Products Section Head

One project, one owner, one escalation path.

Release to a single project. Name the person who fixes bad answers. Route every complaint back into the test set. A pilot with no owner is a demo with a longer life expectancy.



Samy Yusif

AI Products Section Head

Logins are not adoption. Coming back is.

`adoption = repeat users / eligible users`

Measure who returns in week four without being reminded, and how deep they go. Seat counts and prompt counts are the most comfortable way to report activity as though it were value.



Samy Yusif

AI Products Section Head

Measure against the baseline you took before you started.

$$\text{realized} = (\text{base} - \text{actual}) \times \text{rate} - \text{run cost}$$

No baseline means you cannot claim the saving and cannot defend the budget. Same discipline as claiming schedule variance with no baseline programme, and it fails for exactly the same reason.



Samy Yusif

AI Products Section Head

Four slides on architecture, and none of them is a model name.

Which model you pick is the most replaceable decision in the whole study. The layers around it are what you will still be living with in two years.



Samy Yusif

AI Products Section Head